Def. Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to [0, \infty)$ be a non-negative valued.
We said that $\int_I f$ Converges if: f is Riemann-Integrable on any $[c,d] \subseteq I$ with

$$\sup\left\{ \int_c^d f \;\middle|\; [c,d] \subseteq I \right\} < \infty.$$

Prop. If a function $f : [0,\infty) \to [0,\infty)$ is Riemann-Integrable on $[0, A]$ for all $A > 0$,
then $$\int_{[0,\infty)} f = \lim_{A \to \infty} \int_0^A f.$$

Proof. Suppose that the improper integral Converges.
First, given $A > 0$, $\int_0^A f \leq \int_{[0,\infty)} f$ by definition. Since f has only non-negative,
So $\int_0^A f$ increases monotonically, so the limit $\lim_{A \to \infty} \int_0^A f$ exists, with $\leq \int_{[0,\infty)} f$. $\therefore$
Secondly, by the definition and property of supremum, given $\varepsilon > 0$, there is $[c,d] \subseteq [0,\infty)$
Such that $$\int_{[0,\infty)} f - \varepsilon < \int_c^d f \leq \int_{[0,\infty)} f.$$
So, for $A > d$, $$\int_{[0,\infty)} f < \int_c^d f + \varepsilon \leq \int_0^A f + \varepsilon \leq \lim_{A \to \infty} \int_0^A f + \varepsilon.$$
Since $\varepsilon > 0$ was arbitrary, we obtain
$$\int_{[0,\infty)} f \leq \lim_{A \to \infty} \int_0^A f.$$
Consequently, $$\int_{[0,\infty)} f = \lim_{A \to \infty} \int_0^A f. \qquad \boxtimes$$